

Homoscedasticity and Its Testing: From Conditional Expectation to the Breusch–Pagan Test

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Part 10 / 12

**Homoscedasticity: Foundations
and OLS Inference**

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1 Conditional Variance and Homoscedasticity

1.1 Definition of conditional variance

The operator view. Just as $\mathbb{E}[Y | X]$ is a binary operator mapping (X, Y) to a random variable, the conditional variance is a binary operator:

$$\text{Var}(\cdot | \cdot) : (X, Y) \longmapsto \text{Var}(Y | X),$$

where the output is a non-negative random variable on $(\Omega, \mathcal{F}, \mathbb{P})$. Unlike its predecessor, it is defined constructively: we already know $\mathbb{E}[Y | X]$ exists, and we use it to build the new operator.

Construction. Assume $\mathbb{E}[Y^2] < \infty$. Then $(Y - \mathbb{E}[Y | X])^2$ is integrable, and we define:

$$\text{Var}(Y | X) = \mathbb{E}[(Y - \mathbb{E}[Y | X])^2 | X].$$

Setting $u = Y - \mathbb{E}[Y | X]$, this is simply a notational shorthand:

$$\text{Var}(Y | X) = \mathbb{E}[u^2 | X].$$

The output is a $\sigma(X)$ -measurable random variable with values in $[0, \infty)$.

The representing function and its natural domain. By Doob–Dynkin, there exists a Borel function $g : \mathbb{R} \rightarrow [0, \infty)$ such that $\text{Var}(Y | X) = g(X)$ a.s. The function g represents the conditional variance as a function of the realised value of X .

The identity $g(X) = \text{Var}(Y | X)$ is a statement about random variables: it holds $\mathbb{P}$ -almost surely. This means g is only constrained at points x that X hits with positive P_X -probability. More precisely: modifying g on any Borel set B with $P_X(B) = 0$ leaves $g(X)$ unchanged $\mathbb{P}$ -a.s., and hence leaves $\text{Var}(Y | X)$ unchanged. So g is only determined P_X -almost everywhere.

If we wish to extend g to a fully specified function on all of $\mathbb{R}$, the extension on P_X -null sets is arbitrary, subject only to:

- *Borel measurability*: so that $g(X)$ is a well-defined random variable;
- *Non-negativity*: $g(x) \geq 0$ for all x , to preserve the interpretation as a variance.

Remark 1. The range $X(\Omega) \subseteq \mathbb{R}$ and the support $\text{supp}(P_X)$ are different objects and should not be conflated. For instance, X could take every real value (full range) while P_X is concentrated at a single point (if $X = c$ $\mathbb{P}$ -almost surely, even though isolated excursions occur on a $\mathbb{P}$ -null set). Conversely, $X(\Omega)$ could be a proper subset of $\text{supp}(P_X)$ is not possible in general. The correct concept for our purposes is neither: it is the P_X -null sets of $\mathbb{R}$. Any Borel B with $P_X(B) = 0$ is irrelevant for g , regardless of whether B intersects $X(\Omega)$ or $\text{supp}(P_X)$.

Remark 2 (Non-uniqueness of g). The random variable $\text{Var}(Y | X)$ is unique up to a $\mathbb{P}$ -null set on Ω . The representing function g is unique P_X -almost everywhere on $\mathbb{R}$: any two versions agree on every Borel set of positive P_X -measure, and may differ on P_X -null sets.

In particular, the pointwise statement “ $\text{Var}(Y | X = x) = \sigma^2$ for all $x \in \mathbb{R}$ ” is not well-posed: it makes claims about g at points where P_X assigns zero mass, and g is not determined there. The correct formulation is $g = \sigma^2$ P_X -a.e., equivalently $\text{Var}(Y | X) = \sigma^2$ $\mathbb{P}$ -a.s.

1.2 Homoscedasticity

General definition (the Probabilist). Before restricting to regression, consider an arbitrary sequence of random variables $(u_1, \dots, u_n)$ with second moments, defined on a common probability space. No pairing with regressors, no model.

Definition 3. The sequence $(u_1, \dots, u_n)$ is **homoscedastic** if

$$\text{Var}(u_1) = \text{Var}(u_2) = \dots = \text{Var}(u_n).$$

If the sequence has second moments but is not homoscedastic, it is **heteroscedastic**.

The word comes from Greek: *homos* (same) + *skedastikos* (able to scatter), from *skedannynai* (to scatter). The “scedastic” root refers to dispersion or spread; homoscedastic means “scattering equally.”

Regression context: conditional version. In the regression model $Y_i = \beta_0 + \beta_1 X_i + u_i$, the natural generalisation replaces unconditional variances by conditional ones. But there is a subtlety: $\text{Var}(u_i \mid X_i) = g_i(X_i)$ is a *random variable* (it depends on X_i), whereas $\text{Var}(u_i)$ in the general definition is a non-random scalar.

Homoscedasticity in the regression context therefore adds two requirements compared to the general definition:

- (1) Replace unconditional variances by conditional variances: $\text{Var}(u_i) \rightarrow \text{Var}(u_i \mid X_i)$.
- (2) Require that these conditional variances are **non-random** (degenerate): $\text{Var}(u_i \mid X_i) = \sigma^2$ a.s., i.e. the random variable $g_i(X_i)$ collapses to a constant.

Condition (2) has no analogue in the unconditional definition, where $\text{Var}(u_i)$ is already a scalar by definition.

Definition on a sample of pairs. Let $(X_1, Y_1), \dots, (X_n, Y_n)$ be a random vector of pairs defined on a common probability space $(\Omega, \mathcal{F}, \mathbb{P})$. No further assumptions (independence, identical distribution) are imposed yet. For each i , define the conditional variance function

$$g_i(x) = \text{Var}(Y_i \mid X_i = x),$$

so that $\text{Var}(Y_i \mid X_i) = g_i(X_i)$ a.s.

Definition 4. The sample $(X_1, Y_1), \dots, (X_n, Y_n)$ is **homoscedastic** if there exists a constant $\sigma^2 > 0$ such that

$$\text{Var}(Y_i \mid X_i) = \sigma^2 \quad \text{a.s., for each } i = 1, \dots, n.$$

Equivalently, $g_i = \sigma^2$ P_{X_i} -a.e. for each i .

This definition makes no assumption about the relationship between pairs. In particular:

- Different pairs may have different joint distributions (the g_i may be different functions, as long as each equals σ^2 P_{X_i} -a.e.).
- The pairs need not be independent.
- The X_i need not have the same marginal distribution.

Equivalence under identical distribution. Now suppose the pairs are identically distributed: $(X_i, Y_i) \stackrel{d}{=} (X, Y)$ for a single “population” pair (X, Y) . No independence between pairs is assumed. Then all pairs share the same joint distribution, so $g_i = g$ for all i , where $g(x) = \text{Var}(Y \mid X = x)$ is the conditional variance function of the population pair.

Proposition 5. *If the pairs (X_i, Y_i) are identically distributed as (X, Y) , then the sample is homoscedastic if and only if the population pair (X, Y) is homoscedastic, i.e. $g = \sigma^2$ P_X -a.e.*

Proof. Since all $g_i = g$ and all $P_{X_i} = P_X$, the condition $g_i = \sigma^2$ P_{X_i} -a.e. for each i reduces to $g = \sigma^2$ P_X -a.e. □

Remark 6. Independence between pairs plays no role here. It is used later, in Section 2, where we compute the variance of $\hat{\beta}_1$: there the cross-terms $\text{Cov}(u_i, u_j)$ must vanish, and this requires independence (or at least uncorrelatedness) of the errors across pairs. For the equivalence of homoscedasticity definitions, only identical distribution is needed.

Under identical distribution, the sample-level definition collapses to a single condition on the population, and it is natural to state homoscedasticity at the population level:

Definition 7. The pair (X, Y) is **homoscedastic** if

$$\text{Var}(Y \mid X) = \sigma^2 \quad \text{a.s.}$$

for some constant $\sigma^2 > 0$.

Remark 8. In the non-random- X setting of the non-random- X section (Part 1), homoscedasticity was $\text{Var}(\varepsilon_i) = \sigma^2$ for all i . There the x_i are fixed, so each g_i is evaluated at a single point: $g_i(x_i) = \sigma^2$. This is a finite-sample version of the general definition, without any population structure.

Why it matters. The homoscedasticity condition determines whether the OLS estimator has a tractable variance and whether standard inference is valid; see Section 2.

1.3 Are observations i.i.d. under heteroscedasticity? Incidental parameters revisited

We follow Boris's notes of July 3, 2026 (note 57).

The question

Boris conjectured that observations (x_i, y_i) are i.i.d. under H_0 (homoscedasticity) but not under H_a (heteroscedasticity), since under H_a different observations seem to come from distributions with different variances. The Probabilist disagrees: the observations are i.i.d. in either case. Let us sort this out.

What “observation” means

An observation is the pair (x_i, y_i) . We adopt the i.i.d. sampling assumption:

$$(x_i, y_i) \text{ are i.i.d. from the joint law of } (X, Y). \quad (0)$$

The population model is

$$Y = \beta_0 + \beta_1 X + U. \quad (1)$$

The conditional variance function is

$$\sigma^2(x) := \text{Var}(U \mid X = x),$$

and for the i -th observation we write $\sigma_i^2 := \sigma^2(x_i) = \text{Var}(u_i \mid x_i)$.

The Probabilist is right: heteroscedastic observations are still i.i.d.

Under assumption (0), every pair (x_i, y_i) is drawn from the same joint distribution of (X, Y) . Heteroscedasticity means $\sigma^2(x)$ is non-constant in x — but this is a property of the *conditional* distribution of Y given X , not a statement that different observations come from different marginal laws. The joint law of (X_i, Y_i) is the same for every i .

A concrete way to see it: $\sigma_i^2 = \sigma^2(X_i)$ is a function of X_i , which is itself i.i.d. across i . Therefore σ_i^2 is i.i.d. (as a random variable), even though it takes different values at different realisations of X_i .

Remark 9 (Boris’s confusion, unpacked). Boris thought that heteroscedasticity means “observation i comes from a distribution with variance σ_i^2 ,” interpreting i as an index that labels fundamentally different populations. This would indeed break i.i.d. But σ_i^2 is not indexed by i in that sense — it is indexed by the realised value x_i . The subscript i is a shorthand for $\sigma^2(x_i)$, not a label for a separate law. All observations are draws from the same joint (X, Y) distribution; they simply land at different values of X , giving different conditional variances.

What the Probabilist calls “over-relaxed”

Boris initially considered the more general setup where each σ_i^2 could have a *different distribution* depending on i . This would go beyond heteroscedasticity to a model where different observations literally come from different populations. The Probabilist correctly says this is over-relaxed for the present framework: assumption (0) already constrains all pairs to share the same law, so the σ_i^2 must be identically distributed (though not constant).

Homoscedasticity as degeneracy of σ_i^2

In the i.i.d. framework, the Probabilist’s characterisation is sharp:

- **Heteroscedasticity:** $\sigma_i^2 = \sigma^2(X_i)$ is a non-degenerate random variable — it varies with X_i .
- **Homoscedasticity:** $\sigma^2(x) = \sigma^2$ is constant in x , so $\sigma_i^2 = \sigma^2$ a.s. — the random variable $\sigma^2(X_i)$ is *degenerate* (a point mass at σ^2).

Equality across observations then follows automatically from degeneracy: if $\sigma_i^2 = c$ a.s. for all i , then certainly $\sigma_1^2 = \sigma_2^2 = \dots = c$. So in the i.i.d. setting, homoscedasticity *is* the degeneracy of $\sigma^2(X)$ as a random variable.

What about incidental parameters?

Boris asks: should he forget about incidental parameters in this context? The answer depends on which framework we are in.

Random-regressor framework (assumption (0)): No incidental parameters arise. The function $x \mapsto \sigma^2(x)$ is a single unknown object — infinite-dimensional if left unrestricted, but not one-parameter-per-observation. The BP parametrisation $\sigma^2(x) = h(\alpha^\top z)$ imposes a finite-dimensional structure on this function. In this framework, Boris is right to set incidental parameters aside.

Fixed-regressor framework (Koenker’s A.1): Here $x_1, \dots, x_n$ are treated as known constants, not as realisations of a random variable. The conditional variance $\sigma_i^2 = \sigma^2(x_i)$ is then a deterministic unknown number — one per observation. With n free σ_i^2 ’s and n observations, the number of parameters grows with the sample size. This is the incidental parameters problem: no consistent estimator of each σ_i^2 exists because no observation replicates the same x_i . The BP parametrisation resolves this by reducing n free constants to $q + 1$ free constants in α .

Remark 10 (Clarification of an earlier remark). In note 51, when I described the incidental parameters problem, I was speaking in the fixed-regressor sense — each new observation arrives with a fresh σ_i^2 that never gets estimated with more data. This is technically correct in that framework. However, as the Probabilist and Boris established in note 57, the natural framework for the BP test is the i.i.d. random-regressor one (assumption (0)), where the problem does not arise. The incidental parameters language is therefore more relevant to understanding *why a parametric form for $\sigma^2(x)$ is needed*, not to characterising the statistical model per se.

2 Consequences for OLS Inference

We now work in the random- X framework of the random- X framework (Part 1). The data are an i.i.d. sample $(X_1, Y_1), \dots, (X_n, Y_n)$ of pairs, each with the same joint distribution as (X, Y) . The model is

$$Y_i = \beta_0 + \beta_1 X_i + u_i,$$

where $u_i = Y_i - \mathbb{E}[Y_i | X_i]$, so $\mathbb{E}[u_i | X_i] = 0$ by construction. All quantities are random: X_i, Y_i, u_i , and therefore $\hat{\beta}_0, \hat{\beta}_1$ as well.

The OLS estimator is

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_{i=1}^n (X_i - \bar{X})^2} = \frac{S_{XY}}{S_{XX}},$$

where $S_{XX} = \sum (X_i - \bar{X})^2$ and $S_{XY} = \sum (X_i - \bar{X})(Y_i - \bar{Y})$ are now random. We condition throughout on $\mathbf{X} = (X_1, \dots, X_n)$; this is the standard device that recovers the fixed- X analysis as a conditional statement.

2.1 Conditional unbiasedness

$$\mathbb{E}[\hat{\beta}_1 | \mathbf{X}] = \frac{1}{S_{XX}} \sum_{i=1}^n (X_i - \bar{X}) \mathbb{E}[Y_i | \mathbf{X}].$$

By the i.i.d. assumption, (X_j, Y_j) is independent of X_i for $j \neq i$, so $\mathbb{E}[Y_i | \mathbf{X}] = \mathbb{E}[Y_i | X_i] = \beta_0 + \beta_1 X_i$. Substituting:

$$\mathbb{E}[\hat{\beta}_1 | \mathbf{X}] = \frac{\sum_i (X_i - \bar{X})(\beta_0 + \beta_1 X_i)}{S_{XX}} = \beta_1,$$

where the last step uses $\sum_i (X_i - \bar{X}) = 0$. Hence $\hat{\beta}_1$ is conditionally unbiased given $\mathbf{X}$, and therefore unconditionally unbiased: $\mathbb{E}[\hat{\beta}_1] = \beta_1$.

Remark 11 (Why condition on $\mathbf{X}$, not on $(\mathbf{X}, \mathbf{Y})$?). The Probabilist raises a natural question: why do we write $\mathbb{E}[\hat{\beta}_1 | \mathbf{X}]$ and not $\mathbb{E}[\hat{\beta}_1 | \mathbf{X}, \mathbf{Y}]$?

Conditioning on the full sample $(\mathbf{X}, \mathbf{Y})$ makes $\hat{\beta}_1$ degenerate: it is a deterministic function of $(\mathbf{X}, \mathbf{Y})$, so $\mathbb{E}[\hat{\beta}_1 | \mathbf{X}, \mathbf{Y}] = \hat{\beta}_1$ trivially. This is true but says nothing about the model or the parameter β_1 .

Conditioning on $\mathbf{X}$ alone — while leaving $\mathbf{Y}$ random — is the meaningful question: fix the regressor values and ask what $\hat{\beta}_1$ looks like in expectation over all possible response realisations consistent with those X_i . This recovers the classical fixed- X regression analysis as a conditional statement within the random- X framework, and gives a non-trivial result: $\mathbb{E}[\hat{\beta}_1 | \mathbf{X}] = \beta_1$.

In this sense, Y is part of the sample but plays a different role from X : it is the quantity being modelled, not the conditioning variable.

2.2 Variance of $\hat{\beta}_1$ and its relationship to Gauss–Markov

A clarification on purpose (the Probabilist). The derivation of $\text{Var}(\hat{\beta}_1 | \mathbf{X})$ below serves a specific purpose: it shows how homoscedasticity enters the estimation of that variance. However, the Probabilist points out that the conditional variance itself is the central object of the **Gauss–Markov theorem**, not merely a tool for plug-in inference. It is worth keeping these two purposes separate.

Gauss–Markov theorem (statement). Under the conditions of the linear model with $\mathbb{E}[u_i | X_i] = 0$ and homoscedasticity, OLS is **BLUE**: the Best Linear conditionally Unbiased Estimator of β_1 . Here:

- **Linear** means linear in $Y = (Y_1, \dots, Y_n)$. Note: $\hat{\beta}_1 = \sum w_i Y_i$ with weights $w_i = (X_i - \bar{X})/S_{XX}$, so it is linear in Y but *nonlinear* in X .
- **Conditionally unbiased** means $\mathbb{E}[\hat{\beta}_1 | \mathbf{X}] = \beta_1$, as shown in §7.1. This is stronger than unconditional unbiasedness $\mathbb{E}[\hat{\beta}_1] = \beta_1$ (the conditional implies the unconditional by the tower property).
- **Best** means smallest conditional variance among all linear conditionally unbiased estimators.

Linearity in Y arises naturally from MSE minimisation: the objective is quadratic in Y , so the minimiser is linear in Y .

What breaks without homoscedasticity (the Probabilist's guess). Without homoscedasticity, OLS is no longer BLUE — some other linear conditionally unbiased estimator (GLS with weights $1/\sigma_i^2$) achieves smaller conditional variance. What exactly fails: the optimality of OLS within the linear class, not its unbiasedness. Unbiasedness ($\mathbb{E}[\hat{\beta}_1 | \mathbf{X}] = \beta_1$) holds regardless of homoscedasticity, as the derivation in §7.1 did not use it. This is the Probabilist's guess; a full proof would require working through the Gauss–Markov proof with heteroscedastic Ω .

The variance formula. The estimator $\hat{\beta}_1$ is a function of both $\mathbf{X}$ and $\mathbf{Y}$, but its variance $\text{Var}(\hat{\beta}_1)$ is a number — a parameter of the sampling distribution, determined by the model parameters (σ^2 , the distribution of X), not by any particular realisation. Working conditionally on $\mathbf{X}$ gives a tractable intermediate object: the conditional variance $\text{Var}(\hat{\beta}_1 | \mathbf{X})$ is a random variable (it depends on the random $\mathbf{X}$), and the unconditional variance is recovered by taking its expectation via the law of total variance. With $w_i = (X_i - \bar{X})/S_{XX}$ deterministic given $\mathbf{X}$:

$$\hat{\beta}_1 - \beta_1 = \sum_{i=1}^n w_i u_i,$$

and therefore, using conditional independence of u_i given $\mathbf{X}$ (from i.i.d. pairs) and $\text{Var}(u_i | \mathbf{X}) = \text{Var}(u_i | X_i) = g(X_i)$:

$$\text{Var}(\hat{\beta}_1 | \mathbf{X}) = \frac{\sum_{i=1}^n (X_i - \bar{X})^2 g(X_i)}{S_{XX}^2}. \quad (1)$$

This depends on $\mathbf{X}$ (random), with the unknown function g as the only unspecified ingredient.

Remark 12. Three i.i.d. steps justify (1): (i) conditioning on $\mathbf{X}$ makes w_i deterministic; (ii) pairs are i.i.d., so $u_i \perp (X_j, Y_j)$ for $j \neq i$, eliminating cross-terms; (iii) $\text{Var}(u_i | \mathbf{X}) = \text{Var}(u_i | X_i)$ by i.i.d. Each step would fail if i.i.d. were dropped.

2.3 Under homoscedasticity: what is random, what is a parameter

Assume $g = \sigma^2$ (homoscedasticity). The fundamental formula gives

$$\text{Var}(\hat{\beta}_1 | \mathbf{X}) = \frac{\sigma^2}{S_{XX}}.$$

We now parse every symbol carefully.

σ^2 : a parameter. Under homoscedasticity, $\sigma^2 = \text{Var}(u_i | X_i)$ is a non-random constant — a parameter of the model. It is unknown and must be estimated from data.

$S_{XX} = \sum (X_i - \bar{X})^2$: a random variable. The X_i are random, so S_{XX} is a random variable. Conditional on $\mathbf{X}$, it becomes a known scalar.

$\text{Var}(\hat{\beta}_1 | \mathbf{X})$: a random variable. The ratio σ^2/S_{XX} is a random variable (because S_{XX} is random), but with a single unknown scalar σ^2 as its only unspecified ingredient. It is not a parameter — it is a random variable. We do not *estimate* it; we *predict* it.

Estimating σ^2 vs. predicting $\text{Var}(\hat{\beta}_1 | \mathbf{X})$. What software actually does is the following two-step procedure, which mixes estimation and prediction:

1. **Estimate** σ^2 (a parameter) by the unbiased estimator

$$\hat{\sigma}^2 = \frac{1}{n-2} \sum_{i=1}^n \hat{u}_i^2, \quad \hat{u}_i = Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i.$$

One can verify $\mathbb{E}[\hat{\sigma}^2 | \mathbf{X}] = \sigma^2$. Here $\hat{\sigma}^2$ is a statistic — a function of the data, hence a random variable, but an estimator for the non-random parameter σ^2 .

2. **Predict** $\text{Var}(\hat{\beta}_1 | \mathbf{X})$ by substituting the estimate $\hat{\sigma}^2$ for the unknown σ^2 (plug-in principle):

$$\widehat{\text{Var}}(\hat{\beta}_1 | \mathbf{X}) = \frac{\hat{\sigma}^2}{S_{XX}}.$$

This is a random variable (function of data), used as a predictor for the random variable $\text{Var}(\hat{\beta}_1 | \mathbf{X}) = \sigma^2 / S_{XX}$.

By default, statistical software (R, Python, Stata) outputs $\sqrt{\hat{\sigma}^2 / S_{XX}}$ as the “standard error” of $\hat{\beta}_1$. This is valid only under homoscedasticity. Under heteroscedasticity the same formula is used but it predicts the wrong quantity, as discussed in Section 2.4.

Confidence intervals as quantile intervals. The t -statistic

$$t = \frac{\hat{\beta}_1 - \beta_1}{\sqrt{\hat{\sigma}^2 / S_{XX}}}$$

has distribution t_{n-2} conditionally on $\mathbf{X}$ under normality of u and under H_0 . A confidence interval for β_1 is in fact a quantile interval: it inverts the quantiles of the t_{n-2} distribution. Under normality, this is equivalent to a moment-based interval (using the second moment $\hat{\sigma}^2 / S_{XX}$), because the t distribution is fully characterised by its degrees of freedom. Without normality, the asymptotic ($n \rightarrow \infty$) version replaces t_{n-2} by $N(0, 1)$, and the same plug-in formula applies asymptotically.

2.4 Under heteroscedasticity

Suppose $g(X_i) = \sigma_i^2$ varies with i (i.e. g is not constant). The fundamental formula gives

$$\text{Var}(\hat{\beta}_1 | \mathbf{X}) = \frac{\sum_{i=1}^n (X_i - \bar{X})^2 \sigma_i^2}{S_{XX}^2}.$$

This is the scalar sandwich formula. The numerator weights each σ_i^2 by $(X_i - \bar{X})^2$: observations far from $\bar{X}$ contribute more to the variance of $\hat{\beta}_1$.

What happens to $\hat{\sigma}^2$? The quantity $\hat{\sigma}^2 = \frac{1}{n-2} \sum \hat{u}_i^2$ remains computable, but the parameter it estimates no longer exists: there is no single σ^2 in the model. One can show that

$$\mathbb{E}[\hat{\sigma}^2 | \mathbf{X}] \approx \sum_{i=1}^n c_i \sigma_i^2,$$

where weights c_i depend on X_i through the hat matrix. This is a weighted average of the individual σ_i^2 with no interpretable role in the model. The estimand has dissolved.

Consequence for inference. The formula $\hat{\sigma}^2 / S_{XX}$ estimates the wrong object: it targets σ^2 / S_{XX} , whereas the true conditional variance is the sandwich expression above. The two coincide only under homoscedasticity. Under heteroscedasticity:

- $\hat{\sigma}^2 / S_{XX}$ may over- or underestimate $\text{Var}(\hat{\beta}_1 | \mathbf{X})$, depending on the pattern of σ_i^2 relative to $(X_i - \bar{X})^2$.

- The t -statistic does not have its nominal distribution under H_0 ; the test is invalid.
- A consistent estimator of the true conditional variance is the Eicker–Huber–White (HC) estimator:

$$\widehat{\text{Var}}_{\text{HC}}(\hat{\beta}_1 \mid \mathbf{X}) = \frac{\sum_{i=1}^n (X_i - \bar{X})^2 \hat{u}_i^2}{S_{XX}^2},$$

which replaces σ_i^2 by $\hat{u}_i^2$ and is consistent under mild conditions without a homoscedasticity assumption. It is a special case of Newey–West standard errors.

Remark 13. OLS remains conditionally unbiased under heteroscedasticity. The issue is not with $\hat{\beta}_1$ itself but with quantifying its uncertainty. The Gauss–Markov theorem (OLS is BLUE among linear unbiased estimators, conditionally on $\mathbf{X}$) also fails: GLS with weights $1/\sigma_i^2$ is more efficient.

3 Conditional and Unconditional Variance of the OLS Slope Estimate

This section is not directly about the definition of homoscedasticity, but it is central to its application in regression: the variance of the OLS slope estimator $\hat{\beta}_1$, first conditional on the regressors $\mathbf{X} = (x_1, \dots, x_n)$, and then unconditional.

3.1 Setup and the OLS slope estimator

Consider the simple linear model

$$Y_i = \beta_0 + \beta_1 x_i + u_i, \quad i = 1, \dots, n,$$

where the x_i are drawn i.i.d. from some distribution of X , and u_i are errors with $\mathbb{E}[u_i \mid X_i] = 0$ and, under homoscedasticity, $\mathbb{E}[u_i^2 \mid X_i] = \sigma^2$.

The OLS slope estimator is

$$\hat{\beta}_1 = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sum_i (x_i - \bar{x})^2}.$$

3.2 Unbiasedness

Step 1: Conditional expectation of Y_i .

$$\mathbb{E}[Y_i \mid x_i] = \beta_0 + \beta_1 x_i + \mathbb{E}[u_i \mid x_i] = \beta_0 + \beta_1 x_i.$$

Step 2: Conditional expectation of $\bar{y}$.

$$\mathbb{E}[\bar{y} \mid \mathbf{X}] = \frac{1}{n} \sum_i (\beta_0 + \beta_1 x_i) = \beta_0 + \beta_1 \bar{x}.$$

Step 3: Plug in and simplify.

$$\mathbb{E}[\hat{\beta}_1 \mid \mathbf{X}] = \frac{1}{\sum_i (x_i - \bar{x})^2} \sum_i (x_i - \bar{x}) [(\beta_0 + \beta_1 x_i) - (\beta_0 + \beta_1 \bar{x})] = \frac{\beta_1 S_{XX}}{S_{XX}} = \beta_1.$$

By the law of iterated expectations, $\mathbb{E}[\hat{\beta}_1] = \beta_1$: the estimator is unbiased both conditionally and unconditionally.

Unbiasedness of $\hat{\beta}_0$. Since $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$:

$$\mathbb{E}[\hat{\beta}_0 \mid \mathbf{X}] = (\beta_0 + \beta_1 \bar{x}) - \bar{x} \beta_1 = \beta_0.$$

3.3 Conditional variance of $\hat{\beta}_1$

Write $\hat{x}_i = x_i - \bar{x}$ and $S_{XX} = \sum_i \hat{x}_i^2$. Since $\mathbb{E}[\hat{\beta}_1 | \mathbf{X}] = \beta_1$,

$$\text{Var}(\hat{\beta}_1 | \mathbf{X}) = \mathbb{E}[(\hat{\beta}_1 - \beta_1)^2 | \mathbf{X}] = \mathbb{E}[\hat{\beta}_1^2 | \mathbf{X}] - \beta_1^2.$$

Expanding the square in the definition of $\hat{\beta}_1$:

$$\mathbb{E}[\hat{\beta}_1^2 | \mathbf{X}] = \frac{1}{S_{XX}^2} \sum_i \sum_j \hat{x}_i \hat{x}_j \mathbb{E}[(y_i - \bar{y})(y_j - \bar{y}) | \mathbf{X}].$$

We evaluate the inner expectation separately for $i \neq j$ and $i = j$, using the notation $\hat{u}_i = u_i - \bar{u}$, so $y_i - \bar{y} = \beta_1 \hat{x}_i + \hat{u}_i$.

Case 1: $i \neq j$.

$$\mathbb{E}[(y_i - \bar{y})(y_j - \bar{y}) | \mathbf{X}] = \beta_1^2 \hat{x}_i \hat{x}_j - \mathbb{E}[u_i \bar{u} | \mathbf{X}] - \mathbb{E}[u_j \bar{u} | \mathbf{X}] + \mathbb{E}[\bar{u}^2 | \mathbf{X}].$$

Under homoscedasticity and uncorrelated errors: $\mathbb{E}[u_i \bar{u} | \mathbf{X}] = \sigma^2/n$ and $\mathbb{E}[\bar{u}^2 | \mathbf{X}] = \sigma^2/n$. Hence this case gives

$$\beta_1^2 \hat{x}_i \hat{x}_j - \frac{2\sigma^2}{n} + \frac{\sigma^2}{n} = \beta_1^2 \hat{x}_i \hat{x}_j - \frac{\sigma^2}{n}.$$

Case 2: $i = j$.

$$\begin{aligned} \mathbb{E}[(y_i - \bar{y})^2 | \mathbf{X}] &= \beta_1^2 \hat{x}_i^2 + \mathbb{E}[\hat{u}_i^2 | \mathbf{X}] \\ &= \beta_1^2 \hat{x}_i^2 + \sigma^2 - \frac{2\sigma^2}{n} + \frac{\sigma^2}{n} = \beta_1^2 \hat{x}_i^2 + \sigma^2 \left(1 - \frac{1}{n}\right). \end{aligned}$$

Assembling. Apply the algebraic identity $\sum_i \sum_j a_i a_j = (\sum_i a_i)^2$: with $a_i = \hat{x}_i$ this gives $\sum_i \sum_j \hat{x}_i \hat{x}_j = (\sum_i \hat{x}_i)^2 = 0$; with $a_i = \hat{x}_i^2$ it gives $\sum_i \sum_j \hat{x}_i^2 \hat{x}_j^2 = S_{XX}^2$.

Collecting all terms:

$$\mathbb{E}[\hat{\beta}_1^2 | \mathbf{X}] = \frac{1}{S_{XX}^2} \left[\beta_1^2 S_{XX}^2 - \underbrace{\frac{\sigma^2}{n} \left(\sum_i \hat{x}_i \right)^2}_{=0} + \sigma^2 \left(1 - \frac{1}{n}\right) S_{XX} \right] = \beta_1^2 + \frac{\sigma^2(1 - 1/n)}{S_{XX}}.$$

Therefore:

$$\boxed{\text{Var}(\hat{\beta}_1 | \mathbf{X}) = \frac{\sigma^2 \left(1 - \frac{1}{n}\right)}{S_{XX}}}.$$

3.3.1 Simplification using a standard identity

The identity $\sum_i (a_i - \bar{a})^2 = n(\overline{a^2} - \bar{a}^2)$ gives $S_{XX} = n(\overline{x^2} - \bar{x}^2)$, so

$$\text{Var}(\hat{\beta}_1 | \mathbf{X}) = \frac{1}{n} \cdot \frac{\sigma^2}{\overline{x^2} - \bar{x}^2}.$$

3.4 Unconditional variance and L_2 consistency

By the law of total variance:

$$\text{Var}(\hat{\beta}_1) = \mathbb{E}[\text{Var}(\hat{\beta}_1 \mid \mathbf{X})] + \text{Var}\left(\underbrace{\mathbb{E}[\hat{\beta}_1 \mid \mathbf{X}]}_{=\beta_1}\right) = \sigma^2(1 - \frac{1}{n}) \mathbb{E}\left[\frac{1}{S_{XX}}\right].$$

Consistency. If the x_i are i.i.d. with $\text{Var}X > 0$, then by the LLN, $S_{XX}/n \xrightarrow{p} \text{Var}X$, so $S_{XX} \xrightarrow{p} \infty$ and

$$\text{Var}(\hat{\beta}_1 \mid \mathbf{X}) = \frac{\sigma^2(1 - 1/n)}{S_{XX}} \xrightarrow{p} 0.$$

Since $\hat{\beta}_1$ is unbiased, this is equivalent to $\mathbb{E}[(\hat{\beta}_1 - \beta_1)^2 \mid \mathbf{X}] \rightarrow 0$, i.e. *conditional L_2 consistency*. Taking expectations gives unconditional L_2 consistency: $\text{Var}(\hat{\beta}_1) \rightarrow 0$.

Remark 14. The i.i.d. assumption on X is not strictly necessary. It suffices that S_{XX}/n be bounded away from zero in probability (i.e. that $n/S_{XX} \xrightarrow{p} 0$). Under this weaker condition the conditional variance still tends to zero, and L_2 consistency follows.

Remark 15 (What if homoscedasticity fails?). If $\mathbb{E}[u_i^2 \mid x_i] = \sigma_i^2$ varies with i , the conditional variance becomes

$$\text{Var}(\hat{\beta}_1 \mid \mathbf{X}) = \frac{\sum_i x_i^2 \sigma_i^2}{S_{XX}^2}.$$

OLS remains consistent when this ratio tends to zero, but the standard error formula $\hat{\sigma}^2/S_{XX}$ is invalid. Heteroscedasticity-robust (sandwich/HC) corrections are required.

3.5 $\text{Var}X$ vs $\widehat{\text{Var}}X$: plugging in the sample variance

Recall from Section 3 that under homoscedasticity and the i.i.d. assumption,

$$\text{Var}(\hat{\beta}_1 \mid \mathbf{X}) = \frac{1}{n} \cdot \frac{\sigma^2}{\overline{x^2} - \bar{x}^2}.$$

The denominator $\overline{x^2} - \bar{x}^2 = \widehat{\text{Var}}(X)$ is the sample variance of X (without Bessel correction). The question is whether, for large n , we may replace it by the true $\text{Var}X$:

$$\text{Var}(\hat{\beta}_1 \mid \mathbf{X}) \approx \frac{1}{n} \cdot \frac{\sigma^2}{(\text{Var}X)^2}.$$

3.5.1 Almost sure convergence of $\widehat{\text{Var}}(X)$

By the strong law of large numbers, for an i.i.d. sequence $X_1, X_2, \dots$ with $\mathbb{E}X^2 < \infty$:

$$\widehat{\text{Var}}(X) = \overline{X^2} - \bar{X}^2 \xrightarrow{\text{a.s.}} \mathbb{E}X^2 - (\mathbb{E}X)^2 = \text{Var}X.$$

Applying a continuous function (squaring) to both sides preserves almost sure convergence:

$$\widehat{\text{Var}}(X) \rightarrow \text{Var}X \text{ a.s.} \implies (\widehat{\text{Var}}(X))^2 \rightarrow (\text{Var}X)^2 \text{ a.s.}$$

Assuming $\text{Var}X \neq 0$, taking reciprocals is also continuous at $\text{Var}X$, so

$$\frac{\sigma^2}{(\widehat{\text{Var}}(X))^2} \xrightarrow{\text{a.s.}} \frac{\sigma^2}{(\text{Var}X)^2}.$$

Remark 16 (L_2 convergence is not guaranteed). The Probabilist's warning: almost sure convergence of $\widehat{\text{Var}}(X)$ to $\text{Var}X$ does *not* imply L_2 convergence of the ratio $\sigma^2/(\widehat{\text{Var}}(X))^2$. L_2 would require $\mathbb{E}[(\widehat{\text{Var}}(X))^{-4}] < \infty$, which imposes moment conditions on X beyond $\mathbb{E}X^2 < \infty$. The approximation is therefore justified almost surely, but not necessarily in mean square.

3.5.2 The stronger statement with factor n

Without the factor n , both terms in the difference $\text{Var}(\hat{\beta}_1 | \mathbf{X}) - \frac{\sigma^2}{n(\text{Var}X)^2}$ tend to zero individually (each is $O(1/n)$), so their difference trivially tends to zero. The stronger, more informative statement is

$$n \left[\text{Var}(\hat{\beta}_1 | \mathbf{X}) - \frac{\sigma^2}{n(\text{Var}X)^2} \right] = n \left[\frac{\sigma^2}{n(\widehat{\text{Var}X})^2} - \frac{\sigma^2}{n(\text{Var}X)^2} \right] \xrightarrow{\text{a.s.}} 0.$$

This says the two expressions agree to leading order in n , which justifies swapping $\widehat{\text{Var}X}$ for $\text{Var}X$ in the conditional variance formula.

3.5.3 From conditional to unconditional: a general lemma

Earlier we established that conditional unbiasedness is stronger than marginal unbiasedness:

$$\mathbb{E}[\hat{\beta}_1 | \mathbf{X}] = \beta_1 \implies \mathbb{E}[\hat{\beta}_1] = \beta_1.$$

Analogously, by the law of total variance (with the second term vanishing because $\mathbb{E}[\hat{\beta}_1 | \mathbf{X}] = \beta_1$ is constant):

$$\mathbb{E}[\text{Var}(\hat{\beta}_1 | \mathbf{X})] = \text{Var}(\hat{\beta}_1).$$

This is a special case of the following general lemma.

Lemma 17. *Let ξ and η be random variables on a common probability space with $\mathbb{E}\xi^2 < \infty$, and suppose*

$$\mathbb{E}[\xi | \eta] = \mathbb{E}[\xi].$$

Then

$$\mathbb{E}[\text{Var}(\xi | \eta)] = \text{Var}(\xi).$$

Proof. By the law of total variance:

$$\text{Var}(\xi) = \mathbb{E}[\text{Var}(\xi | \eta)] + \text{Var}(\mathbb{E}[\xi | \eta]).$$

The assumption $\mathbb{E}[\xi | \eta] = \mathbb{E}[\xi]$ (a constant) gives $\text{Var}(\mathbb{E}[\xi | \eta]) = 0$, so

$$\mathbb{E}[\text{Var}(\xi | \eta)] = \text{Var}(\xi). \quad \square$$

Remark 18 (Next step: confidence intervals). To construct confidence intervals for the regression coefficients, we need the unconditional variance of $\hat{\beta}_1$. The lemma above, together with the approximation $\text{Var}(\hat{\beta}_1 | \mathbf{X}) \approx \sigma^2/[n(\text{Var}X)^2]$, gives $\text{Var}(\hat{\beta}_1) \approx \sigma^2/[n(\text{Var}X)^2]$ as well — the same formula with true $\text{Var}X$ replacing its sample counterpart.

4 Variance Approximation and Confidence Interval for $\hat{\beta}_1$

4.1 Chain of approximations

We collect the approximations developed in the preceding sections into a single chain.

Approximation 1 (plug in $\text{Var}X$ for the sample variance, a.s. justified):

$$\text{Var}(\hat{\beta}_1 | \mathbf{X}) = \frac{\sigma^2(1 - 1/n)}{S_{XX}} \approx \frac{\sigma^2}{n(\text{Var}X)^2}.$$

Approximation 2 (pass to unconditional variance via the lemma of Section 3.5):

$$\text{Var}(\hat{\beta}_1) = \mathbb{E} \left[\text{Var}(\hat{\beta}_1 \mid \mathbf{X}) \right] \approx \frac{\sigma^2}{n(\text{Var}X)^2}.$$

Approximation 3 (asymptotic normality by CLT; σ and $\text{Var}X$ assumed known for now):

$$\hat{\beta}_1 \approx \mathcal{N} \left(\beta_1, \frac{\sigma^2}{n(\text{Var}X)^2} \right), \quad \text{i.e.} \quad \beta_1 - \hat{\beta}_1 \approx \mathcal{N} \left(0, \frac{\sigma^2}{n(\text{Var}X)^2} \right).$$

The distribution is symmetric about zero.

Under these approximations, the 0.95-quantiles of the pivot $\beta_1 - \hat{\beta}_1$ are $q_{0.025}$ and $q_{0.975}$ of this normal, so

$$P \left\{ q_{0.025} < \beta_1 - \hat{\beta}_1 < q_{0.975} \right\} \approx 0.95.$$

Rearranging (adding $\hat{\beta}_1$ throughout):

$$P \left\{ \hat{\beta}_1 + q_{0.025} < \beta_1 < \hat{\beta}_1 + q_{0.975} \right\} \approx 0.95,$$

giving the approximate 95% confidence interval

$$\text{CI}_{0.95}(\beta_1) \approx \left[\hat{\beta}_1 + q_{0.025}, \hat{\beta}_1 + q_{0.975} \right].$$

4.2 Sources of non-exact equality

Three distinct approximations are layered in the statement above.

1. **Normality assumption.** If $\mathbb{E}[Y \mid X]$ is normally distributed, then the conditional distribution of $\hat{\beta}_1$ given $\mathbf{X}$ is exactly normal. But normality of $Y \mid X$ may not hold; and even when it does, the *unconditional* distribution of $\hat{\beta}_1$ (averaging over the random $\mathbf{X}$) need not be exactly normal. Normality is an asymptotic approximation via the CLT.
2. **Variance approximation.** Even under homoscedasticity, the exact conditional variance is $\sigma^2(1 - 1/n)/S_{XX}$, not $\sigma^2/[n(\text{Var}X)^2]$. Replacing $\widehat{\text{Var}}(X)$ by $\text{Var}X$ is justified almost surely as $n \rightarrow \infty$, but introduces a discrepancy for finite n .
3. **Plug-in of σ and $\text{Var}X$.** The interval above is not yet a statistic: it depends on the unknown parameters σ and $\text{Var}X$. This is the fourth approximation, discussed next.

4.3 The plug-in principle and the fourth approximation

4.3.1 Pivotal quantities and confidence intervals: general framework

Definition 19. Let $\{P_\theta : \theta \in \Theta\}$ be a family of probability measures and let $T = T(X_1, \dots, X_n, \theta)$ be a function of both the sample and the parameter. We call T a **pivotal quantity** (or simply a *pivot*) if its distribution is the same for every P_θ in the family, i.e. the distribution of T does not depend on θ .

Several remarks clarify the scope of this definition.

Remark 20 (Constants are trivial pivots). Any constant c is a pivotal quantity: it depends neither on the sample nor on the parameter, so its distribution (a point mass at c) is obviously parameter-free. Constants are pivots, but degenerate ones — their distribution carries no information and cannot be used to form a confidence interval. The interesting case is a pivot with a *non-degenerate* distribution F .

Remark 21 (Compensation mechanism). The definition does *not* say that T is constant or distribution-free in the trivial sense. Rather, T typically *depends* on θ as a function, but this functional dependence on θ exactly cancels the distributional dependence on θ that the sample $X_1, \dots, X_n$ carries. The parameter enters T in a way that “removes” the parameter from the distribution of T . This compensation is the essential non-trivial content of pivotality.

Remark 22 (No requirement on the relationship to an estimator). The definition says nothing about point estimation: T need not involve an estimator $\hat{\theta}$ at all. It is a separate, additional feature — important in practice — that a pivot is often built from the difference $\hat{\theta} - \theta$ (or a rescaling thereof). This is natural because:

- (i) $\hat{\theta}$ is consistent, so $\hat{\theta} - \theta$ is “small” for large n and concentrates near zero;
- (ii) inverting the event $\{q_{\alpha/2} < \hat{\theta} - \theta < q_{1-\alpha/2}\}$ isolates θ cleanly, giving a CI.

For a **location parameter** θ (i.e. $X - \theta$ has a distribution free of θ), the difference $\hat{\theta} - \theta$ is a canonical pivot. If $\hat{\theta}$ is the sample mean and $X_i \sim \mathcal{N}(\theta, \sigma^2)$ with σ known, then $\bar{X} - \theta \sim \mathcal{N}(0, \sigma^2/n)$ exactly, for every n — a genuine (non-asymptotic) pivot.

When the pivot has a continuous distribution F with quantiles $q_p = F^{-1}(p)$, a $(1 - \alpha)$ confidence interval is obtained by inverting

$$P\left\{F^{-1}\left(\frac{\alpha}{2}\right) < \hat{\theta}_n - \theta < F^{-1}\left(1 - \frac{\alpha}{2}\right)\right\} = 1 - \alpha,$$

rearranging to isolate θ :

$$\text{CI}_{1-\alpha}(\theta) = \left[\hat{\theta}_n + q_{\alpha/2}, \hat{\theta}_n + q_{1-\alpha/2} \right].$$

4.3.2 Monotone transformation of the pivot

If the pivot $\hat{\theta} - \theta$ has a fixed but non-standard distribution, one may apply a monotone bijection g to obtain a better-known pivot $g(\hat{\theta} - \theta) \sim F$. Assuming g is increasing, inverting gives

$$g^{-1}\left(F^{-1}\left(\frac{\alpha}{2}\right)\right) < \hat{\theta} - \theta < g^{-1}\left(F^{-1}\left(1 - \frac{\alpha}{2}\right)\right),$$

and the confidence interval becomes

$$\text{CI}_{1-\alpha}(\theta) = \left[\hat{\theta} + g^{-1}\left(F^{-1}\left(\frac{\alpha}{2}\right)\right), \hat{\theta} + g^{-1}\left(F^{-1}\left(1 - \frac{\alpha}{2}\right)\right) \right].$$

4.3.3 Asymptotic pivot and asymptotic CI

Two distinct concepts appear here and must be kept separate.

Pivotality is a property relative to the measure family at a *fixed* n : a statistic Z_n is pivotal if its distribution is the same for every P in the family, i.e. it does not depend on the unknown parameter θ . This is a statement about one distribution, not about a sequence.

Stabilisation of a distribution (convergence in distribution) is a property of a *sequence*: $Z_n \xrightarrow{d} Z$ means the laws of Z_n converge weakly to the law of Z as $n \rightarrow \infty$. This says nothing about whether any individual Z_n is pivotal.

In the typical situation, $\hat{\theta}_n - \theta$ is *not* pivotal at any fixed n : its distribution depends on θ . A rescaling h_n (e.g. $h_n(x) = \sqrt{n} \cdot x$) may remove this dependence in the limit:

$$h_n(\hat{\theta}_n - \theta) \xrightarrow{d} F,$$

where F does *not* depend on θ and is non-degenerate. This is *asymptotic pivotality*: the rescaled quantity is not pivotal for any finite n , but its limiting distribution is free of θ . As n grows, the dependence on θ becomes negligible, and in the limit disappears entirely.

If $h_n(\hat{\theta}_n - \theta)$ is a genuine pivot at every fixed n — i.e. its distribution is *exactly* F for every n , not merely in the limit — then we have an exact CI for each n :

$$\text{CI}_{n,1-\alpha}(\theta) = \left[\hat{\theta}_n + h_n^{-1}\left(F^{-1}\left(\frac{\alpha}{2}\right)\right), \hat{\theta}_n + h_n^{-1}\left(F^{-1}\left(1 - \frac{\alpha}{2}\right)\right) \right].$$

When the convergence holds only in the limit, the same formula gives an *asymptotic* CI — exact coverage $1 - \alpha$ is achieved only as $n \rightarrow \infty$. This is the most frequent case in practice. The standard choice $h_n(x) = \sqrt{n} \cdot x$ converts consistency at rate $1/\sqrt{n}$ into a standard normal asymptotic pivot.

4.3.4 Approximation 4: plug-in of unknown parameters

Returning to regression. Under the three approximations above,

$$\beta_1 - \hat{\beta}_1 \approx \mathcal{N}\left(0, \frac{\sigma^2}{n(\text{Var}X)^2}\right).$$

This is not yet a pivot: the variance involves the unknown σ and $\text{Var}X$. The standard errors in the interval

$$\hat{\beta}_1 \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n} \text{Var}X}$$

are not observable. To obtain a usable (statistical) interval, we apply the *plug-in principle*: replace σ and $\text{Var}X$ by consistent estimates $\hat{\sigma}$ and $\widehat{\text{Var}}(X)$:

$$\hat{\beta}_1 \pm z_{\alpha/2} \frac{\hat{\sigma}}{\sqrt{n} \widehat{\text{Var}}(X)}.$$

This introduces the fourth approximation. The resulting interval is an asymptotic $(1 - \alpha)$ CI for β_1 , valid under homoscedasticity, i.i.d. X , and the moment conditions that guarantee consistency of $\hat{\sigma}$ and $\widehat{\text{Var}}(X)$.